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Deploy Model Strategy using FastAPI

For this task, I worked on a **Housing Prices** dataset. The main objective was to build a **machine learning model and deploy it using FastAPI** with an **API structure and monitoring system**.

First, I have loaded the dataset in Python and checked the data. There were no missing values or duplicate records. Then, created Graphs for better understanding the data. Also then, encoded the data by converting categorical values like **yes/no** into numbers **0 / 1** so that the model can understand them.

After that, I trained a **Linear Regression model** to predict **house prices** using features like **area, bedrooms, bathrooms, stories, parking, and other factors**. I evaluated the model using metrics such as **R2 score, Mean Squared Error, and Root Mean Squared Error**.

Then, I saved the trained model using **pickle**,

Next, I deployed the model using **FastAPI**. I created API endpoints:

- **/predict** - to predict house price
- **/health** - to check if the API is running
- **/metrics**- for monitoring .

I added condition to show value error if negative values are given in the input. Also ,created try-except to handle errors during prediction. And added handling for validation errors when input data is missing or incorrect.

For monitoring, I have used logging. The system stores input data, prediction output, and response time in a log file. This helps in tracking performance and finding errors if they occur.

The API was tested , can be accessed locally at:

<http://127.0.0.1:8000/docs#/>

The API returned predictions along with response time, which shows that the deployment is working.

